DID YOU KNOW –
LEAK ADAPTATION

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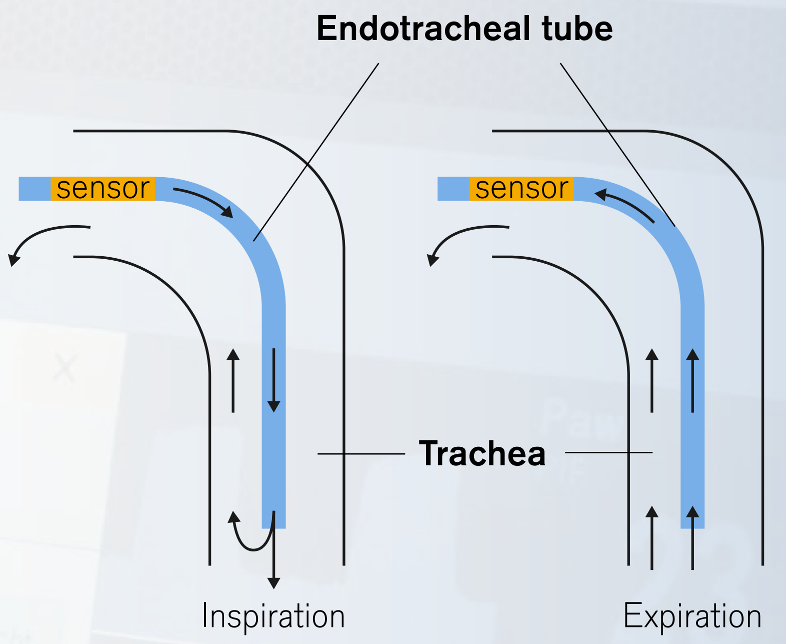
LEAK ADAPTATION

What is it?



During ventilation endo-tracheal tube leakages can occur. A higher percentage of volume is lost during the inspiratory phase than during the expiratory phase because the pressure delivered during inspiratory phase is higher.

Leakage adaptation is an automatic adjustment of trigger and termination criteria according to measured leakage.

Leak adaptation is always active.

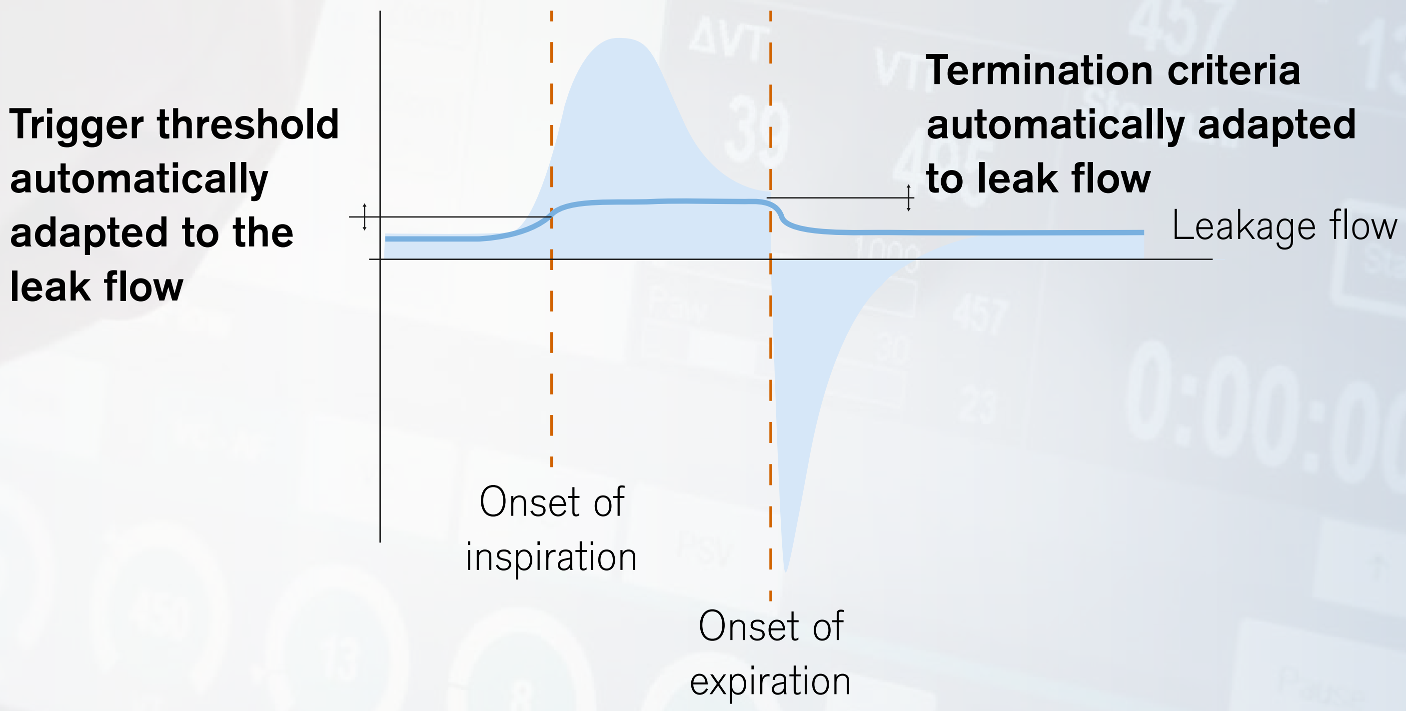
$$\text{Leakage [\%]} = \frac{\text{MVi - MVe}}{\text{MVi}}$$

How is it working?

The leakage flow is automatically subtracted from the total flow in order to determine the patient flow.

After a few breaths the ventilator "learns" the leakage. If the leakage is closed, **the sensitivity of the flow trigger is automatically adapted.**

In all synchronized ventilation modes **the trigger level and the termination criteria are adapted continuously and automatically.**



Why is it helpful to improve outcome?

Leak adaptation (even together with leak compensation) is a powerful combination to decrease the users' cognitive workload and focus on the patient treatment and outcome.

- The user doesn't have to care about trigger correction.
- The inspiratory flow trigger threshold and the inspiratory termination criteria are automatically adapted, and very limited interaction is required from the user.
- It reduces the work of breathing (WOB).
- It shortens the trigger delay time.
- It reduces the rate of auto-cycling.
- Every breath has an optimal management of inspiratory time.